Abhisek Panigrahi

Curriculum Vitae

Education

2012–2016 **B.Tech in Computer Science and Engineering**, *National Institute of Science and Technology*, Berhampur, *8.63*(CGPA).

Projects:

- Movie Recommendation System : The system predicts user rating for a movie based on the past data it has for that user and other users who are related in some way with that user - Implemented Restricted Boltzmann Machine, k-nearest neighbors. Dataset: MovieLens 100k Dataset
- Assembler using C : few basic instructions - main purpose to learn C language
- Presented a seminar on Amazon Echo

Experience

Oct 2018 - Present **Specialist Programmer**, *Infosys*, Bangalore.

Feb 2017 - Sep 2018 **Systems Engineer**, *Infosys*, Bangalore.

Abstract Summarization

Description: Summarize a paragraph

Role: Created torchservable format for `t5-small`, `sshleifer/distilbart-xsum-12-6` and `google/pegasus-xsum`. Added a commit to use the servable format for inference.

Infosys HR model Creation

Description: Create Infosy HR model for Question Answering Task

Role: Fine tuned HR corpus on language model, Trained QA model using the resultant language model, fine tuned squad format hr data on resultant QA model. Reviewed 210 questions from HR data. Augmented, by paraphrasing and by removing unwanted words, HR data to check any improvement on F1.

Question Generation

Description: Generates questions given a paragraph or a document

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Role: Analyzed the quality of the questions generated on an Infosys policy document for unilm and t5 model; recorded the time taken for each model to generate questions. Service endpoint creation for `/paragraph` and `/doc` . Added zip file support. Served t5 model using torchserve: torchserve image creation, archiving to torchservable format and serving. Added containerization support with docker-compose. Migrated flask endpoints to fastapi. Added batch inference for torchservable format. Added aiohttp post request for async support. Recorded benchmark result for fastapi and flask.

Reading Comprehension Platform

Description: Answers questions on uploaded paragraph: highlights the answer in the paragraph

Role: Analyzed different datasets for BERT: How many questions are correct/wrong with some threshold value and how many with no answer. Recorded groundtruth answers for 200 questions on policy documents to compare different models. Tokenization support added for Bert, DistilBert, DistilRoberta. Analyzed 3 documents for answers after generating questions from *Question Generation*. The same is repeated with augmentation(paraphrasing).

FAQ Extractor

Description: Learns similar questions in FAQ dataset(*only text*) and finds out the closest match for asked questions

Role: Contributed in model creation and word embedding creation. Contributed in initial model optimization. Used tensorflow serving to serve the models and stanford glove, Handled multiple service ids, Cached embeddings and models in memory, Created FAQ package, Created docker image for FAQ package, Handled multiple service requests using celery

tools: tensorflow-serving, tensorflow, docker, pyinstaller, python language

Destination Assist

Description: Upon asking the nearby restaurant or coffee shop or any place, the bot will respond back with the top 3 nearby point of interest and will able to help the user with the ratings and whether it is open or close at the moment.

Role: wrote handlers for the chatbot which includes maintaining the context throughout the conversation and switching the context for different kinds of query

tools: Java

Tasks helping team members

Description: Small tasks helping team members

Role: Annotated 20 call transcripts on discovery, connection or objection. Compared 4 different models on question ansawering tasks as correctly answered or not.

Jun 2016 - Jan 2017

Systems Engineer Trainee, *Infosys*, Mysore.

Tennis

It is both command prompt and GUI based. It can: Search a player in the database, Update or Add Player, Book a match, View or Cancel a match, View tournament details

tools: tkinter, cx_Oracle, python language

Certificate and Recognition

- Reinforcement Learning

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- Deep Learning Specialization

August 2018 Most Valuable Player Award VP - Head Infosys Center of Emerging Technology Solutions

Paper Reimplementation

AlexNet paper

VGG paper

Technical Skills

tensorflow	
keras	
python	advanced
git	
c++	
docker	
java	
ubuntu	
C	

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